

Task Manager — Day 2 Progress Notes

Date: 23 September 2026

Day 2 focused on building the user authentication foundation of the Spring Boot Task Manager application, testing registration and login, and saving the completed work in Git and GitHub.

1. What Was Completed

- Created the User entity with id, username, email, and password fields.
- Connected the User entity to MySQL using Spring Data JPA.
- Created UserRepository using JpaRepository.
- Added findByEmail() for login lookup.
- Created UserService for user-related business logic.
- Created AuthController with registration and login endpoints.
- Configured Spring Security to allow registration and login endpoints.
- Added BCrypt password hashing.
- Verified that passwords are stored as BCrypt hashes.
- Created LoginRequest DTO for login input.
- Tested successful login with correct credentials.
- Tested failed login with an incorrect password.
- Initialized Git repository and committed Day 2 changes.
- Created the public GitHub repository and connected the local project to it.
- Renamed the local branch from master to main and pushed the project to GitHub.

2. Project Structure Added

```
src/main/java/com/hitesh/Task_Manager/  
■■■ config/  
■ ■■■ SecurityConfig.java  
■■■ controller/  
■ ■■■ AuthController.java  
■■■ dto/  
■ ■■■ LoginRequest.java  
■■■ entity/  
■ ■■■ User.java  
■■■ repository/  
■ ■■■ UserRepository.java  
■■■ service/  
■■■ UserService.java
```

3. Authentication APIs

Method	Endpoint	Purpose
POST	/api/auth/register	Register a new user

POST	/api/auth/login	Verify email and password
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4. BCrypt Password Hashing

The registration flow was changed so the password is encoded before it is saved. The database therefore stores a BCrypt hash instead of the original plain-text password.

Example database value observed during testing: \$2a\$10\$... (BCrypt hash)

5. Login Testing

Correct credentials

```
Invoke-RestMethod -Uri "http://localhost:8080/api/auth/login" -Method POST -ContentType
"application/json" -Body '{"email":"hitesh2@gmail.com","password":"123456"}'
```

Result: Login successful

Incorrect password

```
Invoke-RestMethod -Uri "http://localhost:8080/api/auth/login" -Method POST -ContentType
"application/json" -Body '{"email":"hitesh2@gmail.com","password":"wrongpassword"}'
```

Result: Invalid email or password

6. Git Progress

Day 1 had already been committed. During Day 2, the new authentication files and pom.xml changes were committed.

Commit	Description
7a8aeb1	Day 1 - Spring Boot project setup
86e171c	Day 2 - User registration and login

7. GitHub Connection

The local repository was connected to the GitHub repository and the local master branch was renamed to main.

Remote: https://github.com/hiteshram18/Task_Manager.git

The project was successfully pushed to GitHub. Final status: branch main is up to date with origin/main and the working tree is clean.

8. Important Security Note for Next Session

Before making further public GitHub pushes, the MySQL database password currently present in application.properties should be moved to an environment variable. This prevents the database password from being exposed in a public repository.

9. Next Step — Day 3

The next planned feature is JWT authentication. The current login only returns a success/failure message. JWT will allow the application to issue a token after successful login and use that token to protect Task Manager APIs.

Planned flow: Register → Login → Verify credentials → Generate JWT → Send token with protected requests → Access Task APIs.

Day 2 checkpoint: User registration, BCrypt hashing, login verification, Git version control, and GitHub backup are working.